

Part 3: MCP & RAG

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Overview

Today we will look at two recent popular ways for LLM's to pull contextual data and other digital artifacts. This data is meant to be used in conjunction with user AI sessions.

- ▶ Model Context Protocol (MCP)
- ▶ Retrieval-Augmented Generation (RAG)

Review From last Time

Part 1: Prompting the LLM

We looked at how to prompt and why it is important.

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Part 2: LLM Tokenization

We looked at how the LLM breaks our conversations down into discrete “chunks” known as tokens. We discussed ways to improve AI sessions and increase efficacy of session results.

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Model Context Protocol: Architecture and Components

Model Context Protocol

- ▶ The MCP creates a standardized, two-way connection for AI applications, allowing LLMs to easily connect with various data sources and tools.
- ▶ MCP builds on existing concepts like tool use and function calling but standardizes them. This reduces the need for custom connections for each new AI model and external system.
- ▶ MCP enables LLMs to use current, real-world data, perform actions, and access specialized features not included in their original training.

MCP Host

The LLM is contained within the MCP host, an AI application or environment such as an AI-powered IDE or conversational AI. This is typically the user's interaction point, where the MCP host uses the LLM to process requests that may require external data or tools.

MCP Client

The MCP client, located within the MCP host, helps the LLM and MCP server communicate with each other.

It translates the LLM's requests for the MCP and converts the MCP's replies for the LLM. It also finds and uses available MCP servers.

MCP Server

The MCP server is the external service that provides context, data, or capabilities to the LLM.

It helps LLMs by connecting to external systems like databases and web services, translating their responses into a format the LLM can understand which helps developers provide diverse functionalities.

MCP Transport Layer

The transport layer uses JSON-RPC 2.0 messages to communicate between the client and server, mainly through two transport methods:

- ▶ Standard input/output (stdio): Works well for local resources, offering fast, synchronous message transmission
- ▶ Server-sent events (SSE): Preferred for remote resources, allowing efficient, real-time data streaming

Retrieval-Augmented Generation: Architecture and Components

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is an approach that improves how language models respond by connecting them to external data sources.

My Lab Example

I kept “starting over” by accident because my models did not have a set of static reference materials.

I started creating a PDF document to give my models a “brain” and allow for a common starting point.

Training Resources

Relevant training materials.